

Counterfactuals and their temporal asymmetries

PHIL UA-78 Metaphysics

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Holding the past fixed

Another pervasive pattern

Typically, when we are considering a counterfactual whose antecedent is about some fixed interval of past time and whose consequent is about events during and later than that interval, we implicitly “hold fixed” facts about how things went before that time.

If I had done such-and-such, I would have avenged a wrong.

If I had skipped the Wordle today, that would have been the first time I did so in months.

If this video were downloaded 10,000 times, that would set a new record for us.

“Backtracking” contexts

There are some special contexts—Lewis calls them “backtracking” contexts—where we interpret counterfactuals in a distinctive way that allows for widespread variations in the past. One example is this sentence from Jackson:

- ▶ If I had jumped out of this tenth-floor window, that would have been because someone had put a safety net in place.

Consequents like ‘that would have been because. . .’ often suggest this kind of treatment. Also, counterfactuals whose consequent is *about* earlier times often push us towards a backtracking reading, since they would be boring under a ‘standard’ reading.

Lewis sets backtracking readings aside in this paper.

The need for a 'transition period'

Even if we confine ourselves to “standard” contexts, we must be careful not to overstate the generality of the ‘holding history fixed’ pattern.

Right up to t , the match was stationary and a foot away from the striking surface. If it had been struck at t , would it have travelled a foot in no time at all? No; we should sacrifice the independence of the immediate past to provide an orderly transition from actual past to counterfactual present and future. (Lewis CDTA 463)

Lewis calls this a ‘transition period’.

The puzzle of counterfactuals under determinism

Last week, I defined determinism as the following claim:

Determinism

All truths are necessitated by the laws of nature together with the complete truth about the intrinsic state of the world during any initial interval of history.

We can also put this by saying that all truths are “historically necessary” at every time.

Variant definitions of determinism

Van Inwagen uses a stronger definition:

Instant-Determinism

All truths are necessitated by the laws of nature together with the complete truth about the intrinsic state of the world at any instant.

There is some reason to think that even paradigms like Newtonian Mechanics do not obey Instant-Determinism: *velocities*, understood as derivatives of position, are not intrinsic to an instant.

However, as far as I know, all the live candidate physical theories that imply Determinism also imply

Interval-Determinism

All truths are necessitated by the laws of nature together with the complete truth about the intrinsic state of the world during any interval of nonzero temporal length.

The puzzle of counterfactuals under determinism

Suppose determinism is true. Both of the following seem tempting:

Laws If you hadn't come to class today, every actual law of nature would have been true. [Cf. Carroll's 'SC']

Past If you hadn't come to class today, the intrinsic character of the first billion-year stretch of the history of the universe after the Big Bang would have been exactly the same.

But the following principle looks non-negotiable:

Closure If a state of affairs is necessitated by some states of affairs each of which would have obtained if you hadn't come to class today, then it would have obtained if you hadn't come to class today.

So given determinism, Laws, and Past, we have to say that if you hadn't come to class today you would have come to class today!

For 'standard' contexts, Lewis gives up Laws. He thinks that in deterministic worlds, if things had been different at one time, a 'miracle' would have occurred shortly before that time.

- ▶ NB: That does not mean that false propositions can be laws! The so-called 'miracle' is something inconsistent with the *actual* laws, which wouldn't all have been laws if you hadn't come to class.
- ▶ (What would the laws have been then? Maybe they'd be exception-tolerant variants of the actual laws.)

The alternative choice

One could instead keep Laws (and SC), and reject Past (for ordinary contexts, in deterministic worlds).

Does this require giving up on the general idea that we can hold propositions about earlier times fixed when evaluating 'standard' counterfactuals about later times?

No! We can say that if things had been different at t , the course of history substantially earlier than t would have been only *slightly* different, whereas the course of history after t would have been significantly different.

Lewis considers this possibility, and expresses a worry:

Indeed, it is hard to imagine how two deterministic worlds anything like ours could possibly remain just a little bit different for very long. There are altogether too many opportunities for little differences to give rise to big differences. (Lewis CDTA 45).

Lewis's worry about preserving Laws

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I don't see much to worry about, since all the live candidates for deterministic laws are *continuous*. That means that when possible histories start out sufficiently close, they can stay close for arbitrarily long before the differences blow up to the macroscopic scale.

Lewis's closeness relation

Analysis 1

Lewis sets himself the task of giving an *analysis* of 'If A it would be that C ' on its standard reading. As a foil, he starts with:

Analysis 1

Where A is entirely about affairs during t_A , "If it were that A , then it would be that C " is true iff C is true at every possible world w such that:

- (1) A is true at w ;
- (2) w is exactly like the actual world at all times before a transition period beginning shortly before t_A ;
- (3) w conforms to the actual laws of nature at all times after t_A , and
- (4) during t_A and the preceding transition period, w differs no more from our actual world than it must to permit A to hold.

Why isn't this good enough?

Lewis makes the following complaints about Analysis 1:

- ▶ It is silent about the truth or falsity of counterfactuals whose antecedents are *not* entirely about affairs in a stretch of time.
- ▶ It doesn't fit into a general semantic framework which shows what all the different possible interpretations of counterfactuals have in common.
- ▶ Aside from the 'transition period', it incorrectly rules out backwards counterfactual dependence as absolutely impossible, even under weird laws of nature (e.g. laws that allow for time travel).
- ▶ It just helps itself to the contrast between past and future, ruling out any non-circular explanation of that in terms of counterfactuals, and leaving a need to explain why we have this concept, rather than a different concept in which the roles of 'past' and 'future' are reversed.

Instead, Lewis offers:

Analysis 2

A counterfactual 'If it were that A , then it would be that C ' is ... true if and only if some ... world where both A and C are true is more similar to our actual world, overall than is any world where A is true but C is false.

This is just what we had on Monday, but with 'more similar' where I earlier wrote 'closer'.

Spelling out 'similar'

Fine's objection: some worlds where Nixon presses the button and there is still no nuclear holocaust are more similar to the actual world than any worlds where Nixon presses the button and there is a nuclear holocaust, so the analysis incorrectly predicts that 'If Nixon had pressed the button there would have been a nuclear holocaust' is false.

Lewis's Response: the sense of 'similar' under which Analysis 2 is true is not the familiar one under which Fine's similarity claims are correct.

Remaining task for Lewis: say what it takes for one possible world to be more similar to the actual world than another, in the relevant way.

- (1) It is of the first importance to avoid big, widespread, diverse violations of law.
- (2) It is of the second importance to maximize the spatiotemporal region throughout which perfect match of particular fact prevails.
- (3) It is of the third importance to avoid even small, localized, simple violation of law.
- (4) It is of little or no importance to secure approximate similarity of particular fact, even in matters that concern us greatly.

Why is this supposed to work?

Lewis's claim: Assume that the actual world is deterministic, and that Nixon's button is wired straight up, at t , to the nuclear missile launcher. Then, by this measure of similarity, the most similar worlds to the actual world where Nixon presses the button at t are ones which match actuality perfectly up to shortly before t , then diverge via a small miracle (i.e. a violation of the actual laws of nature), then proceed according to the actual laws of nature, in such a way that the nuclear holocaust ensues.

Why is this supposed to work?

Five worlds:

- w_0 The actual world: Nixon doesn't press button.
- w_1 Actual past up to shortly before t ; small miracle; button pressed; nuclear holocaust; no more miracles.
- w_2 No miracles; button pressed; entire past very different.
- w_3 Actual past up to shortly before t ; small miracle; button pressed; small miracle in wire; signal doesn't reach launcher; no holocaust.
- w_4 Actual past up to shortly before t ; small miracle; button pressed; big miracle wipes out all of the traces of his having done so; actual future from soon after t ; hence no holocaust.

w_4 is further from w_0 than all the others because of clause (1).

w_2 is further from w_0 than w_1 or w_3 because of clause (2).

w_3 is further from w_0 than w_1 because of clause (3).

Worlds with the same future and distant pasts

Lewis is trying for something very ambitious: his definition of the similarity relation does not contain any words like 'future' and 'past'.

This leads to a worry about worlds that match actuality in the future but not the past, and only have one small miracle. (See Elga, 'Statistical Mechanics and the Asymmetry of Counterfactual Dependence').

One option for Lewis to get around this would be to change clause (2) so that only match in *backwards-closed* regions counts for anything. (That would mean giving up on the aspiration to give the analysis without building in any time-asymmetry.)

I am given the opportunity to bet that a coin will land heads. I decline. It's tossed and lands heads. I say 'If I had bet, I would have won'. Is this true?

- ▶ Version 1: the coin-toss is outside my future light cone.
- ▶ Version 2: if I had bet, the air currents around the coin would have been a little different. . .
- ▶ Version 3: if I had bet, I would have been the one to toss the coin.

Pollock's stolen coat problem

Lewis's truth conditions make 'If my coat had been stolen last year, it would have been stolen on December 31st' come out automatically true (assuming that on any given day, it would only have taken a small miracle).

This seems like a bad result!